**DS 4420: Machine Learning and Data Mining 2** **(Due: Feb. 23 2026)**

# Homework Assignment #4

*Professor:* Eric Gerber *Name:* ______________________

**Instructions**: Please include the following information on the first page of your completed homework write-up:

1. Your name
2. DS 4420
3. Homework #4

You will submit **up to four files** to Gradescope for this homework:

- Handwritten/latex typeset answers may be included in their own **.pdf** file or as part of either of the below
- A **.ipynb** file with all python code/output
- A **.pdf** file knitted from an included **.rmd** with all R code/output

Answers that are not supported by reasoning/work will not receive full credit. **Homework is due by 11:59 pm, via Gradescope, on the date above.** Late submissions will **not** be accepted, but you may receive extra credit for early submission (see syllabus for details).

**You will also be graded on organization/neatness of the submitted files.**

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**Stock Data Time Series (50 points)**

**(1: 10 points)** In the next couple of problems, you will build a model to predict the future value of a stock (of your choice) in **python**. To complete the assignment, you will need the **yfinance** module. Install it via pip:

```
pip install yfinance
```

**NOTE:** Any results from this assignment should probably not be used to make investment decisions...

**(a)** Choose a stock to analyze and enter it as a string to the **yf.Ticker()** function, then save its values for the past year in a data frame. This website lists some trending stock tickers and allows you to search for others. For example, Dr. Gerber used the ticker **GK**:

```
GK = yf.Ticker("GK")
GK_values = GK.history(start="2025-02-10")[['Close']]
```

After reading in the stock data:

- Make a time series plot to visualize the full series of the stock over the last year
- Discuss **in detail (at least a couple sentences)** based on the plot your impression of the time series and if you think there is a strong trend and/or seasonality to the data, and what that would mean for the analysis of these data with basic time series approaches.

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**(b)** Plot out the autocorrelation function (ACF) and partial autocorrelation function (PACF) for the stock time series data. Discuss **in detail (at least a couple sentences)** what the plots suggest to you about which modeling strategy will be appropriate (choose between $AR(p)$, $MA(q)$, both, or neither).

**(2: 40 points)**

**(a)** Prepare the data for analysis by splitting the data into 80% training set and 20% test set. Fit either an $AR(p)$ or $MA(q)$ model using either the **AutoReg()** (for $AR(p)$) or **ARIMA()** (for $MA(q)$) functions from the **statsmodels** package. Define your value of $p/q$ based on the ACF/PACF plots from the previous part (b). Then:

- Print out the **.summary()** and/or the **abs(model.roots())** (if you used $AR(p)$) of the model.
- Mention whether the stationarity assumption seems to be met or violated (if you used $AR(p)$) and (regardless of model) check the residual diagnostic plots and discuss them.
- Interpret what the first couple coefficients of the model mean practically.
- Predict the values in the test set using the model and visualize the performance with a plot.
- Calculate the $MAE$ and $RMSE$ for the model performance on the test set.

**(b)** Fit the opposite model (either $MA(q)$ or $AR(p)$, depending on what you just did). Define your value of $p/q$ based on the ACF/PACF plots from earlier. Repeat the five steps from (a).

**(c)** Try using the **ARIMA()** function to fit a combined $ARMA(p, q)$ model or (especially if you think that the stationarity assumption is violated) a $ARIMA(p, d, q)$ model. Define your value(s) of $p$ and/or $q$ based on the ACF/PACF plots from earlier. If you use the integration filter, set it to $d = 1$. Repeat the five steps from (a).

**(d)** Make a choice concerning which of the models ($AR(p)$, $MA(q)$, or some combination) seems to do the best job of predicting your stock of choice and discuss **in detail (at least a few sentences)** what the choice of model implies about the stock data, if you think the model is worth using and/or if needs more work and, if it does, what you would change/update.

**Google Trends Data (35 points)**

**(3: 15 points)** In the next couple of problems, you will build a model to predict trends in a Google search term of your choice.

**(a)** Go to Google Trends and enter a search term of your choice (no more than two words, for simplicity: for example, Dr. Gerber used "baseball"). Under **Explore search trends**, set the timeframe to **2004-Present**, then click the **Download CSV** button in the **Interest over time** cell. Rename this file to something more convenient (e.g. "baseball_trends.csv") and save it in your working directory.

**(b)** Load the data into **python** and make a time series plot to visualize the full series. Discuss **in detail (at least a couple sentences)** based on the plot your impression of the time series and if you think there is a strong trend and/or seasonality to the data.

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**(c)** Plot out the ACF and PACF for the search term time series data. Discuss **in detail (at least a couple sentences)** what the plots suggest to you about which modeling strategy will be appropriate (choose between $AR(p)$, $MA(q)$, both, or neither).

**(Extra Credit (5 points))** Load the data into **R** and make the same plots as in (b) and (c). You do not have to provide the discussion again.

**(4: 20 points)**

**(a)** Prepare the data for analysis by splitting the data into 90% training set and 10% test set. Fit at least two different candidate time series models (of your choice, among $AR(p)$, $MA(q)$, $ARMA(p,q)$, and/or $ARIMA(p,d,q)$) to the Google trends search term data in **python**. Discuss whether the data appear to be stationary, check the residual diagnostic plots, provide some general interpretation of the coefficients, and visualize predictions and assess predictive accuracy using the test set.

**(b)** Based on the candidate models you fit in (a), determine which one seem to work best for predicting the Google trends search term. Discuss **in detail (at least a few sentences)** what the choice of model implies about the data, if you think the model is worth using and/or if it needs more work and, if it does, what you would change/update.

**(Extra Credit (5 points))** Complete the programming parts of (a) and (b) in **R**. You do not have to provide the discussion parts again.

## Time Series Problem Ideation (15 points)

**(5)** Discuss a real data problem and identify a corresponding data set, which you think would be appropriate to analyze with Time Series approaches. The data set **must not** be a commonly used/standard time series data set (e.g., stock prediction or any data sets that are commonly used in tutorials/online examples: the TAs and Dr. Gerber will be checking). Provide:

**(a)** A working link to the data set or to the website(s) from which the data set may be collected and a description of the problem of interest and how the data could be used to answer the problem with Time Series analysis.

**(b)** A discussion of any ethical concerns there might be in using the data and/or using the results of the model.

**(c)** Your intuition as to why a Time Series approach would be appropriate for this problem.